

POKHARA UNIVERSITY

Level: Bachelor	Semester: Fall	Year: 2017
Programme: BE		Full Marks: 100
Course: Mobile and Wireless Communication		Pass Marks: 45
		Time: 3 hrs

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Compare features of 2G and 3G mobile communication. Mention the evolution path from 1G upto 3G. 8
b) Define Frequency reuse, cluster, co-channel and adjacent (channel interference in cellular system. Find the frequency reuse factor and the cluster size that should be used for maximum capacity. There are six co-channel cells in the first tier and all of them are at the same distance from the mobile. (Assume minimum SIR = 20 dB and path loss exponent, $n = 4$). 7
2. a) What is cellular system capacity? "The number of channels assigned to a cell eventually becomes insufficient to support the required number of users" if you agree present any two common technique with brief explanation to provide more channels per unit coverage area. 8
- b) Define Reflection, Diffraction and Scattering. If 100 W is applied to a unity gain antenna with 900 MHz carrier frequency, find the received power in dBm at a free space distance of 100m from the antenna. What is P_r (20 km)? (Assume unit gain receiver antenna.) 7
3. a) Sketch the waveforms and block diagram to explain BPSK transmission for a binary sequence 101100 modulated under BPSK technique. 8
b) Describe outdoor propagation model of Okumura. 7
4. a) What do you mean by Spread Spectrum modulation technique? Explain Direct Sequence Spread Spectrum (DSSS) and Frequency Hopped Spread Spectrum (FHSS) modulation. 8
b) Draw architecture of Long-Term Evolution. Mention functions of its 7

- components. 3
5. a) Explain the fundamentals of equalizations. Explain with block diagram the function of Rake receiver. 3
 - b) Explain Space Division Multiple Access (SDMA) and Hybrid Spread Spectrum Multiple Access Techniques used in wireless communication. 7
 6. a) The most remarkable features in GSM is on-the-air privacy, justify how it is made possible" Also, explain the data bases used in GSM architecture. 7
 - b) Design a syndrome calculator for a (7,4) cyclic Hamming code generated by the polynomial $G(D)=D^3+D+1$. Evaluate the syndrome for $Y=(1000101)$. 8
 7. Write short notes on: (Any two) 2x5
 - a) Trunking and Grade of Service (GOS)
 - b) Co-channel and Adjacent channel Interference
 - c) FDMA and TDMA

POKHARA UNIVERSITY

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Attempt all the questions.

1. a) Briefly explain the two most attractive features of 3G compared with 2G. 7
b) What are the two most predominant forms of interferences in cellular telephone system? Briefly explain them. 8
 2. a) What is cellular system capacity? "The number of channels assigned to a cell eventually becomes insufficient to support the required number of users" if you agree present any two common techniques with explanation to provide more channels per unit coverage area. 7
b) If a transmitter produces 50W of power, express the transmit power in units of 8
 - i. dBm
 - ii. dBW.
- If 50W is applied to an unity gain antenna with a 900 MHz carrier frequency, determine the received power in dBm at a free space distance of 100m from the antenna. What is P_R (10 Km)? Assume unity gain for the receiver antenna.
3. a) Briefly explain the importance of coherent bandwidth and coherent time in small scale multipath fading. 8
b) "Second generation cellular systems, are completely digital" is this true? If yes, then give any six reasons behind digitization. 7
 4. a) What is GMSK modulation? Explain the GMSK transmission system along with the functionalities of two main systems of GMSK modulation, CDPD and Mobitex. 8
b) Find a generator polynomial $g(x)$ for a (7,4) cycle code and find code vectors for the following data vectors: 1010 and 1011. 7

5. a) What is diversity, angle diversity and path diversity? 6

OR

Briefly explain space diversity technique with functional block diagram.

b) A spread spectrum communication system is characterized by the following parameters: 9

Duration of each information bit, $T_b = 4.095 \text{ ms}$

Chip duration of a PN sequence, $T_c = 1 \mu\text{s}$

Determine the processing gain and jamming margin if $(E_b/N_0) = 10$

and the average probability of error $P_e = 0.5 \times 10^{-5}$.

6. a) The most remarkable features in GSM is on-the-air privacy, answer how is it made possible? Also explain the 3-different data bases used in GSM architecture. 8

b) How is LTE different from WiMAX in the 4G world of technology? 7

7. Write short notes on: (Any two) 2×5

a) Hata Model

b) RAKE Receiver

c) Forward CDMA channels

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Attempt all the questions.

1. a) Compare CDMA, GSM & DECT system. 8
b) Derive the relation $N = i^2 + j^2 + j^2$, where N is a cluster size. 7
 2. a) What is handoff and mention its types? Explain different types of channel assignment strategies used in cellular system. 8
b) If a Transmitter produces 50W of power, express the power in units of
i. dBm 8
ii. dBW 7
- If 50W is applied to an antenna with a 900 MHz carrier frequency, find the received power in dBm at a free space distance of 100m from the antenna. Assume unity gain for both transmitting and receiving antenna.
3. a) Explain Hata model. 8
b) Discuss about GSMK transmission and reception with necessary diagrams. 7
 4. a) Explain MSK and derive the expression for MSK signal as a special type of FSK signal. 8
b) Compare block coding with convolution coding and explain convolution coding system with the help of suitable diagram. 7
 5. a) Explain in detail the various factors to determine the algorithm for adaptive equalizer. 8
b) Compare FDMA and TDMA technology. 7
 6. a) Explain Frequency Hopped Multiple Access (FHMA). 7
b) Describe forward and reverse channel parameters of IS-95 CDMA. 8
 7. Write short notes on: (Any Two) 2×5=10
a) Doppler Spread and Coherent time
b) Comparison of merits and demerits of 3G and 4G wireless standards
c) Diversity

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Semester -- Spring
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Attempt all the questions.

1.
 - a) Define Wireless Communication. Explain its evolution in brief. 8
 - b) Why hexagonal cells geometry are used in the design of cellular system. Prove that the co-channel reuse ratio is given by $Q = \sqrt{3}N$, where $N = i^2 + ij + j^2$ is the cluster size. 7
2.
 - a) Explain how cell splitting and sectoring improves the coverage and capacity in Cellular System. 7
 - b) A City with coverage area of 2000 Sq Km is covered with a 12-cell system each with a radius of 1.45 Km. If the total spectrum allocated is 38.5 MHz with a full duplex channel bandwidth of 25 MHz. Assume GOS of 0.02 for a blocked calls cleared system is specified and the offered traffic per user is 0.03 Erlangs and traffic intensity of each cell is 84 Erlang, Compute
 - i) Number of Cells in the Service area.
 - ii) Number of Channels per cell.
 - iii) Maximum Carrier Traffic.
 - iv) Total number of users that can be served for 2% GOS.
3.
 - a) What are the role of large scale and small scale propagation model in support to Okumura model? 8
 - b) Briefly explain the Rayleigh and Ricean fading distribution. 7

a) Briefly explain different types of spread spectrum modulation technique. Discuss the principle of OFDM modulation Scheme. 3

b) Why do you need speech coding technique? Explain the basic concept of VOCODERS. 7

a) Why Equalization and Diversity is needed in Wireless communication. Explain the training and tracking modes of operation for Adaptive Equalizer in detail. 3

b) What is Multiple Access Technique? Compare TDMA, FDMA and CDMA. 7

a) Briefly explain the architecture of GSM system and the role played by HLR, VLR EIR, OMC and AUC in GSM. 3

b) Explain the term Spread Spectrum. What are the main features of CDMA Technology? 7

7. Write short notes on (Any Two) : 2x5

a) DECT.

b) QPSK

c) LTE.

POKHARA UNIVERSITY

Level: Bachelor	Semester: Fall	Year : 2019
Programme: BE		Full Marks: 100
Course: Mobile and Wireless communication		Pass Marks: 45
		Time : 3hrs.

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Attempt all the questions.

1. a) Give short description on:
 - i. Need for developing cellular mobile telephone system.
 - ii. Examples of 2.5 G and 3G standards.
 - iii. SMS.
- b) Explain the basic concept of Frequency reuse. What are the two most predominant forms of interferences in cellular communication illustrate.
2. a) Find the median path loss using Okumura's model for $d=50$ km, $h_{te}=100$ m, $h_{re}=10$ m, in a urban environment if the base station transmitter radiates an EIRP of 1 kw at a carrier frequency of 900MHz. Find the power at the receiver. (Okumura's analysis curve will be provided)
- b) Do the following as stated.
 - i. Define slow and fast fading.
 - ii. What is coherence bandwidth?
 - iii. What are the effects of fading?
 - iv. Define effective aperture.
3. a) Determine the number of Full-Duplex channels available in a cluster and the total capacity for a cellular system where there are 10 clusters, each consisting of 20 cells with 18 channels in each cell

- b) the most remarkable feature in GSM is on-the-air privacy. answer how it is made possible? Also, explain the 3-different data bases used in GSM architecture. 7
4. a) Consider hamming codes that belong to the family of (n, k) block error correcting codes and for minimum distance and $m=3$, give the properties of hamming codes. 5
- b) Draw the offset QPSK and Standard QPSK diagrams. Also show their signal states and possible transmission between the states. 5
- c) Compare BASK, BFSK and BPSK for the variables Bandwidth, Performance and bit rate. Determine minimum Bandwidth for a BPSK modulator with carrier frequency of 50 MHz and an input bit rate of 500kbps. 5
5. a) Sketch the structure of OFDM transmitter and Receiver section and hence briefly explain the OFDM mechanism. 8
- b) Define ISI. Draw and explain the concept of discrete time linearly equalized communication system that should implement algorithm to completely reduce ISI distortion. 7
6. a) What are Reverse CDMA and Forward CDMA channel types? Explain. 7
- b) Compare and Contrast TDMA and FDMA technologies based on its working principle and applications. 8
- Write short notes on (Any two):
- Link Budget design
 - Soft and Hard Hand-off
 - LTE and its role in communication
- 2.5

POKHARA UNIVERSITY

Level: Bachelor	Semester: Spring	Year 2019
Programme: BE		Full Marks: 100
Course: Mobile and Wireless Communication		Pass Marks: 45
		Time 3hrs

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) Explain the two most important features of 4G compared to 3G mobile communication. 7
b) Discuss the cell geometry. Why hexagonal cells are used for planning of cellular system. Prove that the co-channel reuse ratio is given by $Q = \sqrt{3N}$, where the cluster size $N = i^2 + ij + j^2$ and relate with the concept of frequency reuse. 8
2. a) A cellular service provider decides to use a digital TDMA scheme which can tolerate an signal to interference ratio of 15dB in the worst case. Find the optimal value of N for (a) Omni-directional antennas. (b) 120° sectoring, and (c) 60° sectoring. Should sectoring be used? If so, which case (120° or 60°) should be used? Assume a path loss exponent of $n=4$ and consider trunk efficiency. 8
b) Define EIRP and Effective Aperture, and hence deduce the free space equation relating them. 7
3. a) Assume a receiver is located 10 Km from a 50 W transmitter. The carrier frequency is 900 MHz, free space propagation is assumed, $G_t=1$, and $G_r=2$, find (a) the power at the receiver, (b) the magnitude of the E-field at the receiver antenna (c) the rms voltage applied to the receiver input assuming that the receiver antenna has a purely real impedance of 50Ω and is matched to the receiver. 8
b) Write the waveforms to explain QPSK transmission for a binary sequence 101100 modulated under QPSK technique. 7
4. a) What do you mean by Hamming code? Construct an even parity 7-bit Hamming code when a bit word 1010 is to be transmitted. 8
b) What are the advantages of equalization technique? Compare linear and Non-Linear equalization. 7

a) Explain the techniques behind multiple access. Compare FDMA, TDMA techniques. 3

b) The world is already moved towards the development phase of 6th Generation in wireless communication, on this regard, what could be your expectations on the trend of Development of ICT in Nepal. 7

a) What are the services and features of GSM? Construct GSM architecture and Explain in detail. 3

b) Sketch and explain well labeled GSM codec, showing all processing units and all required equipment, to get coded speech signal from a input speech. 7

1. Write short notes on: (Any two) 2x5

a) Link Budget Design

b) Hand-off and its types

c) LTE

POKHARA UNIVERSITY

Level: Bachelor	Semester: Fall	Year : 2020
Programme: BE		Full Marks: 100
Course: Mobile and Wireless Communication		Pass Marks: 45
		Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) Compare and contrast CDMA, GSM and DECT wireless system. 7
 b) If a signal to interference ratio of 15dB is required for satisfactory forward channel performance of a cellular system, what is the frequency reuse factor and cluster size that should be used for maximum capacity if path loss exponent is (a) $n=4$, (b) $n=3$. Assume that there are 6 co-channels cells in the first tier, and all of them are the same distance from the mobile. 3
2. a) What is radio wave propagation? Explain basic propagation mechanisms. 1+0
 b) Calculate the mean path loss using Okumura's model for $d = 50 \text{ km}$, $h_t = 100 \text{ m}$, $h_r = 10 \text{ m}$ in a suburban environment. If the base station transmitter radiates an $EIRP$ of 1 kW at a carrier frequency of 900 MHz , find $EIRP$ (dBm) and the power at the receiver where gain at receiving antenna is 10 dB . 3
3. a) What is frequency Reuse in cellular system? How does the idea is helpful in the improvement of system capacity? Explain with the various techniques used. 3
 b) Why M-ary modulation is important? Explain QPSK modulation and demodulation. 2+5
4. a) The generator polynomials of a $1/3$ binary convolution encoder is given by:

$$g_1(x) = x^3 + x^2 + 1$$

$$g_2(x) = x^3 + x$$

$$g_3(x) = x^3 + 1$$
 Then calculate the encoded bit sequence for input sequence 011. 3
 b) What is the advanced feature of non-linear equalizer over linear 2

- equalizer? Explain the blocks of one non-linear equalization technique.
5. a) Explain the techniques behind multiple access. Compare TDMA and CDMA techniques. 8
- b) How is LTE different from WiMAX in 4G technology. 7
6. a) Explain the logical channels used in GSM system with the help of GSM call. 7
- b) What is Channel coding? Explain CRC technique with an example. 4+4
7. Write short notes on: (Any two) 2×5
- a) Flat fading
- b) Hard and soft hand-off
- c) OFDM

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2021

Programme: BE

Full Marks: 100

Course: Mobile and Wireless Communication

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

- a) What are the advantages and disadvantage of cellular system? Explain the evolutions of wireless networks. 8
- b) For a seven cell reuse pattern. Find the reuse factor of the minimum distance between centres of co-channel cells is 18 km. Radius of cell in 3 km and the distance between adjacent cells in the seven cell pattern in 6 km. 7
- a) What is co-channel and adjacent interference? If a signal-to-interference ratio of 15 dB is required for satisfactory forward channel performance of a cellular system, what is the frequency reuse factor and cluster size that should be used for maximum capacity if the path loss exponent is (a) $n = 4$, (b) $n = 3$? Assume that there are six co-channel cells in the first tier, and all of them are at the same distance from the mobile. Use suitable approximations. 8
- c) Discuss the indoor propagation models for wireless communications. 7
- i) Why link budget analysis is needed? Evaluate the feasibility of a 4 km link, with one access point and one client radio. The access point is connected to an antenna with 12dBi gain, with a transmitting power of 20 dBm and a receive sensitivity of -90dBm. The client is connected to an antenna with 10dBi gain, with a transmitting power of 18dBm and a receive sensitivity of -80dBm. The cables in both systems are short, with a loss of 3dB at each side at the 5.1 GHz frequency of operation. 7
-) Discuss the generation and reception techniques of BPSK and QPSK, modulation techniques. Point out difference between BPSK and QPSK. 8

What is cyclic code? Explain convolutional encoder with suitable illustration. 2

What do you mean by vocoder? Explain the types of vocoders. 7

What is the role of equalizer? Discuss the blocks of adaptive equalization technique with one example. 7

Why multiple access technique is necessary? Differentiate between FDMA, TDMA, and CDMA. 8

Write down the major features of LTE. Point out the difference between DECT, WLAN, WiFi, WiMAX. 3

Enlist the basic features and services of GSM. Draw and explain GSM system architecture. 7

Write short notes on: (Any two) 2/3

- a) Difference between Android and ios
- b) Reverse CDMA Channel
- c) Frequency hopping